

Understanding Lymphatic Anatomy and Abnormalities at Imaging

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Abbreviations: GBCA = gadolinium-based contrast agent, GLA = generalized lymphatic anomaly, GRE = gradient echo, GSD = Gorham-Stout disease, ISSVA = International Society for the Study of Vascular Anomalies, LAM = lymphangioma, LM = lymphatic malformation, PLE = protein-losing enteropathy, 3D = three dimensional

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Explain the most common anatomic variants for the central lymphatic system.
- Describe the role of various imaging modalities including nuclear medicine lymphoscintigraphy, conventional lymphangiography, and MR lymphangiography in the diagnosis and treatment of lymphatic disorders.
- Discuss the range of lymphatic disorders including focal lesions such as LM, generalized lymphatic conditions, and acquired lymphatic disorders.

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Lymphatic abnormalities encompass a wide range of disorders spanning solitary common cystic lymphatic malformations (LMs) to entities involving multiple organ systems such as lymphangioma, myomatosis. Many of these disorders are rare, yet some, such as secondary lymphedema from the treatment of malignancy (radiation therapy and/or lymph node dissection), affect millions of patients worldwide. Owing to complex and variable anatomy, the lymphatics are not as well understood as other organ systems. Further complicating this is the variability in the description of lymphatic disease processes and their nomenclature in the medical literature. In recent years, medical imaging has begun to facilitate a deeper understanding of the physiology and pathologic processes that involve the lymphatic system. Radiology is playing an important and growing role in the diagnosis and treatment of many lymphatic conditions. The authors describe both normal and common variant lymphatic anatomy. Various imaging modalities including nuclear medicine lymphoscintigraphy, conventional lymphangiography, and MR lymphangiography used in the diagnosis and treatment of lymphatic disorders are highlighted. The authors discuss imaging many of the common and uncommon lymphatic disorders, including primary LMs described by the International Society for the Study of Vascular Anomalies 2018 classification system (microcystic, mixed, and macrocystic LMs; primary lymphedema). Secondary central lymphatic disorders are also detailed, including secondary lymphedema and chylous leaks, as well as lymphatic disorders not otherwise easily classified. The authors aim to provide the reader with an overview of the anatomy, pathology, imaging findings, and treatment of a wide variety of lymphatic conditions.

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Introduction

The lymphatic system performs multiple important functions in the human body, including maintenance of fluid balance, absorption of fat from the digestive tract, and immune surveillance for infection and cancer. The lymphatics extend throughout the body, ultimately connecting to the central venous system. Yet, despite the critical importance and extensive presence throughout the body, lymphatics are often overlooked on many imaging examinations owing to small vessel size and variable anatomy, and the methods for imaging lymphatic tracts and lymphatic disease processes remain underdeveloped. With the emergence of new imaging techniques, including MR lymphangiography, as well as a renewed interest in lymphatic treatment approaches, our understanding of the radiologic appearances of this system and the pathologic processes that may affect this system is growing (1–3).

TEACHING POINTS

- The thoracic duct is the largest lymphatic channel in the body, collecting lymph from everywhere in the body except the right upper hemithorax, right upper extremity, and right side of the neck.
- In the lungs, lymphatics are present in the interlobular septa and subpleura and along the bronchovascular structures.
- Both the peripheral and central lymphatics can be imaged by using MRI. Two basic sequences are used: a heavily T2-weighted acquisition and precontrast and dynamic postcontrast breath-hold 3D fat-saturated T1-weighted gradient-echo (GRE) acquisitions after the administration of a GBCA.
- The ISSVA classifies the various lymphatic malformations into common cystic LM, GLA, GSD, channel-type LM (also known as central conducting lymphatic anomaly), acquired progressive lymphatic anomaly, and primary lymphedema.
- Several approaches may be used depending on the clinical scenario and underlying pathologic condition, including selective embolization, percutaneous or catheter-based sclerosis, or more novel techniques such as thoracic duct stent placement.

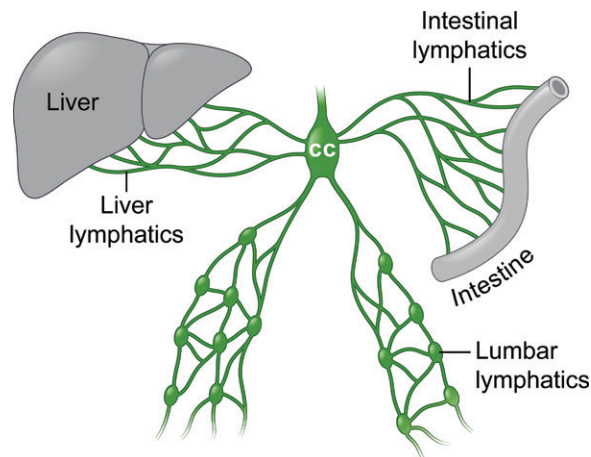


Figure 1. Illustration shows lymphatic drainage from the abdomen and lower extremities. The liver, intestinal, and lumbar lymphatics drain into the cisterna chyli (CC), a focal dilated lymphatic vessel, which becomes the thoracic duct. The way in which the lymphatics join to drain into the CC is highly variable.

This article describes the anatomy and physiologic role of the lymphatic system and discusses some of the major conditions outlined in the International Society for the Study of Vascular Anomalies (ISSVA) 2018 classification system (4). Additional pathologic processes not included in the ISSVA classification and secondary central lymphatic disorders are also described. The various modalities available for imaging evaluation of lymphatic pathologic conditions and imaging-based treatment strategies conclude this overview of lymphatic disorders.

Lymphatic Physiology and Anatomy

Lymphatic vessels allow fluid balance, absorption of digested fats, and immune surveillance. Small lymphatic capillaries join to form lymphatic vessels that drain into filtering lymph nodes and then into central lymphatics, ultimately draining into the venous system.

The central lymphatic anatomy is highly variable. In general, lymphatics from the pelvis and lower extremities along with abdominal lymphatics from the bowel and liver drain into the cisterna chyli (Fig 1). Chyle (a milky odorless fluid) is a mixture of chylomicrons (absorbed from the intestines) and clear lymph from the pelvis and lower extremities. The cisterna chyli is located behind the right diaphragmatic crus at the T12–L2 level (Fig 2) and gives rise to the thoracic duct. It is often identifiable at cross-sectional imaging as an ovoid structure with fluid attenuation or signal intensity and may be mistaken for a lymph node, especially when large. The characteristic location, fluid attenuation or signal intensity, and connection to a tubular structure superiorly (thoracic duct) can help differentiate it from a pathologic finding.

The thoracic duct is the largest lymphatic channel in the body, collecting lymph from everywhere in the body except the right upper hemithorax, right upper extremity, and right side of the neck. Most commonly, the thoracic duct enters the thorax through the aortic hiatus and is located between the azygos vein and aorta. Once in the thorax, it crosses to the left at the T5–T7 level and then ascends to the level of C7. It finally drains into the venous system, posterior to the left internal jugular vein at its junction with the left subclavian vein, commonly referred to as the left venous angle (Fig 3). However, the anatomy of the thoracic duct is highly variable, with a single trunk reported in only 16% of cadavers (5). Multiple additional anatomic variants have been described (Fig 3B) (6). In patients with congenital heart disease, who may have abnormal situs, anatomic variability in the appearance of the thoracic duct and cisterna chyli is more common (7).

A smaller right lymphatic duct drains the right hemithorax, right upper extremity, and right side of the neck. It most commonly begins with the union of the right jugular, right subclavian, and right bronchomediastinal ducts, but again, this anatomy is highly variable (8). Mirroring the thoracic duct, the right lymphatic duct typically joins the venous system at the confluence of the right subclavian vein and right internal jugular vein (Fig 3).

In the lungs, lymphatics are present in the interlobular septa and subpleura and along the bronchovascular structures (9). Disease processes that affect the lymphatics can therefore lead to thickening of the interlobular septa and peribronchovascular interstitium. From here, lymphatics coalesce and drain to the hila and then into the mediastinal lymphatics (Fig 4). In comparison,



Figure 2. (A, B) Axial (A) and coronal (B) CT images show the cisterna chyli (arrow), which is located posterior to the right diaphragmatic crus between the levels of T12 through L2. (C) Axial T2-weighted MR image shows that the cisterna chyli (arrow) is hyperintense.

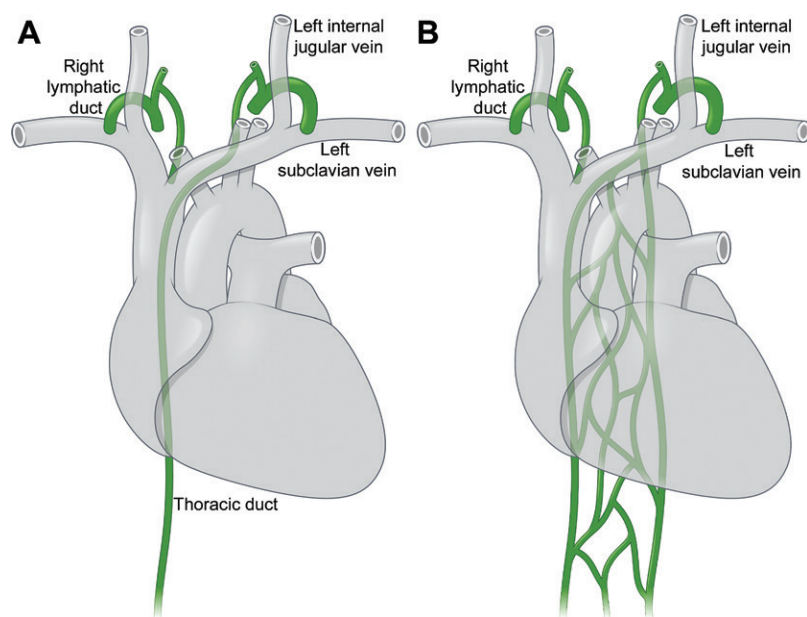


Figure 3. Illustrations show the most common thoracic duct anatomy. (A) The thoracic duct anatomy is highly variable, with the most commonly described variation including a single duct ascending on the right, crossing midline between T5 and T7, and then emptying into the venous system at the confluence of the left internal jugular vein and left subclavian vein. (B) Another common variant is two ducts (a left and a right duct) with multiple anastomoses ascending in the thorax ultimately joining on the left side of the thorax before draining into the left-sided central veins.

fluid in the pleura drains through stoma on the parietal pleura to the lymphatics along the ribs (10). Anteriorly, these lymphatics drain to the internal mammary chain and posteriorly to the intercostal lymphatics. The stoma of the parietal pleura along the diaphragm can directly drain into the mediastinum, retrosternal lymphatics, and celiac lymphatics.

Imaging Evaluation of Lymphatic Flow Abnormalities

Nuclear Medicine: Lymphoscintigraphy

Lymphoscintigraphy can be performed to help detect abnormal lymphatic flow. In this procedure, a technetium-99m-labeled probe, often sulfur colloid, is injected in the subcutaneous tissues between the toes of both lower extremities. This radiotracer is taken up by the lymphatics, and serial images are acquired to assess the progression of the radiotracer in the lymphatic system by

using a γ camera (11). Abnormal movement of radiotracer can be seen in a variety of conditions including lymphedema, chyle leak, etc (Fig 5). This technique is similar to sentinel lymph node mapping except that the injection site in sentinel node mapping is near the cancerous lesion.

Conventional Lymphangiography

In one common method to perform conventional lymphangiography, the interventional radiologist first performs intranodal or pedal lymphangiography (usually with ethiodized oil) to help identify the cisterna chyli. Oil-based contrast agents are viscous and radiopaque and allow effective targeting of the cisterna chyli. When performed for assessment of chyle leak, this procedure is often therapeutic as well as diagnostic; ethiodized oil frequently leads to an intralymphatic inflammatory reaction, which may by itself result in chyle leak cessation (12). A drawback to the use of an oil-based contrast agent for lymphangiography is the

risk of symptomatic embolization to the lungs after draining into the venous system (Fig 6). Patients who undergo lymphangiography will usually have some ethiodized oil in the pulmonary circulation, although this is not clinically significant in low doses. For this reason, high volumes of oil-based contrast agents should be avoided in patients, especially those with underlying pulmonary disease such as chronic obstructive pulmonary disease (COPD). In practice, a dose of no more than 20 mL is typically used (13). Also, systemic arterial embolization can occur in patients with right-to-left shunt with devastating complications, and hence this procedure is relatively contraindicated in this scenario (14).

Conventional liver lymphangiography can be performed by positioning a needle under US guidance in the periportal space of a branch of the left or right portal vein. Opacification of the liver lymphatics can then be performed by the injection of a water-soluble contrast agent (1,15). A similar technique has also been used to opacify the mesenteric lymphatics by transabdominally placing a needle adjacent to an anterior superior mesenteric vein or mesenteric lymph node under US guidance (2).

MRI Techniques

Both the peripheral and central lymphatics can be imaged by using MRI. Two basic sequences are used: a heavily T2-weighted acquisition and precontrast and dynamic postcontrast breath-hold three-dimensional (3D) fat-saturated T1-weighted gradient-echo (GRE) acquisitions after the administration of a gadolinium-based contrast agent (GBCA). Generally, the heavily T2-weighted sequence covers the anatomy from the cervical to inguinal regions. This allows the depiction of larger central lymphatics and generalized edema (dermal, mesenteric, and mediastinal) and the localization and severity of lymphedema, ascites, and pleural effusion. This sequence also helps in the identification of cystic lymphatic malformations (LMs). It has been shown that T2-weighted imaging can be used to map lymphatic abnormalities to help predict outcomes and hospital stays in single-ventricle palliation (16). However, smaller lymphatics within organs and in the extremities frequently cannot be resolved with this sequence (17). Furthermore, T2-weighted imaging does not provide information relating to lymphatic flow and cannot localize regions of active chyle leak; for such information, dynamic postcontrast T1-weighted images must be obtained.

Dynamic postcontrast imaging of the lymphatics may be performed upstream of an injection site after intracutaneous (3,18,19) or intranodal (17,18) injection of GBCA allow-

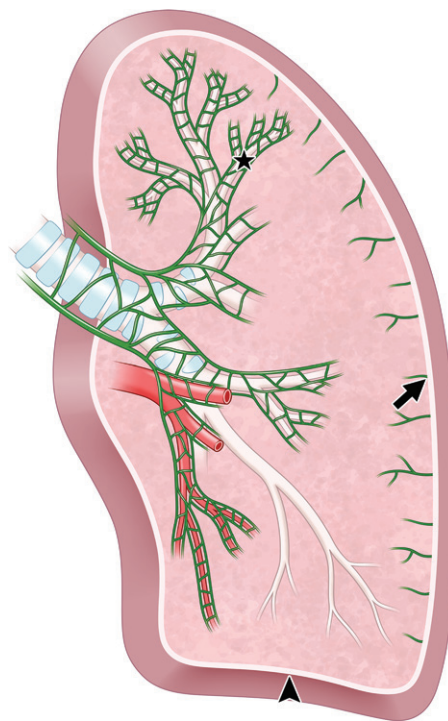


Figure 4. Illustration shows the lymphatic anatomy in the lungs. Lymphatics (green lines) run along the interlobular septa (arrow), in the sub-pleura (arrowhead), and along the bronchovascular structures (★). These drain into the hilum and then the mediastinum. Pleural contents are drained through the parietal pleura to the chest wall, abdominal, and mediastinal lymphatics.

ing noninvasive assessment of lymphatic flow, detection of potential chyle leaks, and mapping of small lymphatic channels in the extremities (Fig 7). Intracutaneous injection is usually performed in the web spaces of the feet (or hands if upper extremity imaging is desired) and is commonly used for peripheral lymphatic imaging in patients with lymphedema, although some authors have described modifications to adapt this to imaging of the central lymphatic system as well. Usually with intracutaneous injections in the dermis, the contrast agent does not extend to the central lymphatics. Hence, if central lymphatic imaging with intracutaneous injection is desired, then one of the adaptations is to encourage patient activity like walking, knee bends, etc. Some special contrast agents have also been described, although these are experimental or not clinically available. Transnodal injection is usually performed after insertion of a catheter into a lymph node (most commonly an inguinal lymph node) under US guidance. Angiocatheters are typically chosen for lymph node catheterization because the metallic needle can be removed, leaving an MRI-safe plastic catheter that may remain in situ for the duration of imaging.

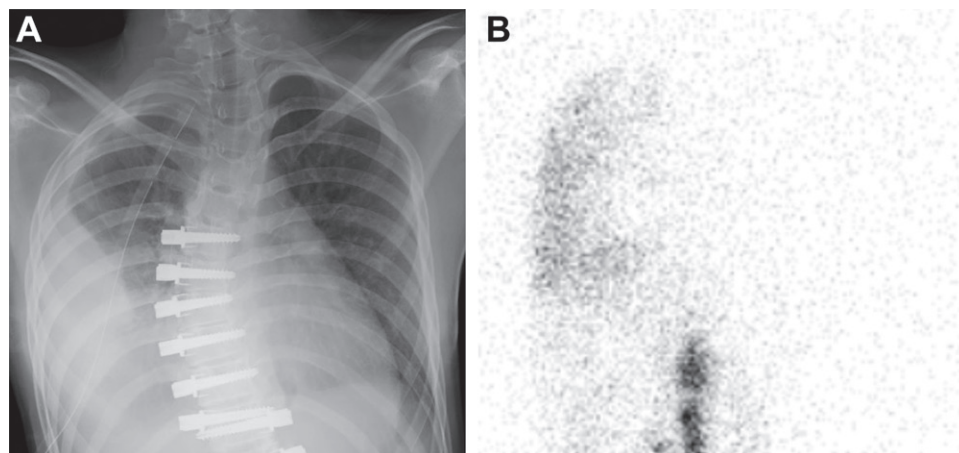


Figure 5. Active chyle leak in a 13-year-old girl with a history of scoliosis and thoracic duct embolization who was admitted for recurrent high-output chylous pleural effusion. **(A)** Frontal chest radiograph shows a moderate right pleural effusion after chest tube placement. **(B)** Nuclear medicine lymphoscintigraphy was performed. Frontal planar image shows radiotracer accumulating diffusely in the right pleural space at the 2-hour imaging, consistent with active chyle leak. Another thoracic duct embolization was performed, with resolution of the pleural effusion.



Figure 6. Pulmonary embolization of an oil-based contrast agent from lymphangiography in a 58-year-old man with worsening dyspnea. Axial noncontrast CT image shows hyperattenuating material in the lungs due to pulmonary embolization of oil-based contrast agent from lymphangiography. High volumes of contrast agents should be avoided in patients with underlying pulmonary diseases. Patients with known right-to-left shunt are at risk for systemic embolization.

Appropriate positioning can be confirmed by injection of a water-soluble contrast agent under fluoroscopy or by injecting saline under US guidance and watching the lymph node increase in size. Needle position has also been confirmed by using contrast-enhanced US to directly visualize flow into the lymphatics (20). Alternatively, positioning can be confirmed with US alone, ensuring that the tip of the angiocatheter is near the corticomedullary junction to avoid capsular rupture of the lymph node with saline injection. GBCA transits in the lymphatics rapidly, much faster than ethiodized oil does during conventional lymphangiography. Therefore, intranodal injection

is performed with the patient in the MRI suite so that dynamic imaging may commence immediately after injection.

When well-performed, intranodal GBCA injection results in striking enhancement of the central lymphatics and thoracic duct with little to no venous contamination in the early postcontrast acquisitions. The contrast agent typically passes to the retroperitoneal lymphatics approximately 2 minutes after injection, progressing to the cisterna chyli at 3–6 minutes, and finally opacifying the thoracic duct to the left venous angle at 5–9 minutes (17).

Similar to conventional lymphangiography techniques, central lymphatic MRI can be performed following intrahepatic and intramesenteric GBCA injections, with the former showing benefit in the diagnosis of protein-losing enteropathy (PLE). Access to the lymphatics is performed by using US guidance and confirmed at fluoroscopy with a water-soluble contrast agent. GBCA is then injected with the patient in the magnet, at a similar rate as with intranodal injection, but the dynamic imaging is more frequent given that the transit time is more rapid. With intrahepatic MR lymphangiography, the contrast agent typically flows centrally to the liver hilum flowing to the cisterna chyli and thoracic duct. During intramesenteric MR lymphangiography, the contrast agent opacifies the mesenteric lymphatics before flowing to the cisterna chyli and thoracic duct (2,21).

Conversely, imaging the peripheral lymphatics following intracutaneous injections usually results in some degree of venous enhancement, particularly near the site of injection. Consequently, some authors recommend administering ferumoxytol, an iron-based blood pool agent, before

intracutaneous administration of GBCA to null signal caused by intravascular uptake of GBCA when attempting to image lymphatics within an extremity (22,23). Some operators use gadobenate dimeglumine for lymphatic imaging on the supposition that its weak albumin-binding properties will improve lymphatic enhancement. However, other authors have achieved success with other group II GBCAs as well (18,24). Sample protocols for nodal and intracutaneous lymphangiography are provided (Tables 1, 2). It should be noted that contrast MR lymphangiography is an off-label use of an MRI contrast agent.

Lymphatic Disorders

There is a wide spectrum of lymphatic disorders ranging from congenital malformations to other processes leading to lymph and/or chyle accumulation or lymphatic flow abnormalities. These are summarized in Table 3. The ISSVA classifies the various LMs into common cystic LM, generalized lymphatic anomaly (GLA), Gorham-Stout disease (GSD), channel-type LM (also known as central conducting lymphatic anomaly), acquired progressive lymphatic anomaly, and primary lymphedema (Table 4) (4). We also discuss some lymphatic diseases not classified in this system (eg, lymphangioliomyomatosis [LAM]) and various other conditions such as chyle leaks and impaired lymphatic flow that can manifest as chylothorax, chylopericardium, chylous ascites, chyluria, plastic bronchitis, and secondary lymphedema.

When fluid collection in body cavities is identified, its chylous nature is diagnosed on the basis of fluid analysis results demonstrating triglyceride levels greater than 110 mg/dL or by the presence of chylomicrons. To differentiate a chylous effusion from a pseudochylothorax or cholesterol effusion, a cholesterol level may be obtained with chylous effusions having a ratio of cholesterol to triglycerides of less than 1. The loss of chyle from the lymphatic system results in electrolyte alterations, nutritional deficiency, and immune suppression. Therefore, identification and treatment of the underlying lymphatic abnormality is crucial (25). Given the heterogeneous conditions that can lead to chylous fluid accumulation, the precise workup and imaging manifestations depend on the underlying disorder and clinical context, for example trauma, malignancy, or multisystem involvement in GLA and GSD. Imaging allows the identification of the leak or obstruction leading to the extralymphatic accumulation of chylous fluid and assessment of anatomic lymphatic variations for pretreatment planning, especially when considering interventional approaches like embolization or stent placement. Likewise, although diagnosis of lymphedema is based on



Figure 7. Dynamic contrast-enhanced MR lymphangiography during intranodal contrast agent injection in a 10-year-old boy with a history of double-outlet right ventricle after transannular patch repair, protein-losing enteropathy (PLE), and liver fibrosis. Initially, bilateral inguinal lymph nodes were accessed under US guidance. Coronal spoiled 3D gradient-echo (GRE) T1-weighted high-resolution imaging with volumetric excitation (THRIVE) dynamic MR image shows lymphatic ducts in the retroperitoneum confluent with a cephalad small thoracic duct (arrow). The thoracic duct courses along the left side of the spine and joins the venous system, likely at the confluence of the left jugular and subclavian veins.

clinical assessment, the role of imaging in surgical and interventional planning is growing.

Lymphatic Malformations

Common Cystic LMs

The common cystic LMs in the 2018 ISSVA classification system are subdivided into macrocystic, microcystic, and mixed types (4). These lesions are defined as low-flow vascular malformations due to abnormal development of lymphatic channels (26). They do not communicate with normal lymphatics and typically manifest in childhood. Often these lesions are transspatial, crossing through anatomic planes, and are best evaluated by using cross-sectional imaging (CT and MRI) that can provide a 3D perspective of the extent of these lesions.

The macrocystic type typically occurs below the mylohyoid and contains cysts larger than 1–2 cm (27). US findings include multiple anechoic

Table 1: Sample Protocols for Central Nodal MR Lymphangiography*

Sequence	Notes
3D heavily T2 weighted Coronal fat-suppressed T2 weighted; additional planes such as axial can also be obtained as needed	Allows depiction of larger central lymphatics and localization of ascites and pleural effusion
Coronal dynamic fat-suppressed T1-weighted MR lymphangiography	Dynamic acquisition simultaneous with intranodal injection, with multiple phases approximately every minute for 15 min

Note.—Contrast-enhanced MR lymphangiography is an off-label use of an MRI contrast agent.

*This technique is commonly used for evaluation of central lymphatics. MRI preparation consists of inserting a catheter into a lymph node and confirming the position.

Table 2: Sample Protocols for Peripheral Intracutaneous MR Lymphangiography*

Sequence	Notes
3D coronal T1-weighted fast field echo	Echo time (TE) selection sequence allows the radiologist or technologist to select the optimal TE to suppress signal from the gadolinium in blood (20)
Postgadolinium multistation coronal 3D Dixon	Isotropic voxels are preferred; use optimal TE as chosen by the radiologist
Coronal 3D T2-weighted fat-suppressed turbo spin echo	Heavily T2-weighted to depict the severity and distribution of lymphedema
Repeat multistation coronal 3D Dixon	Second lymphangiography acquisition
MR angiography/MR venography sequence (3D Dixon)	Select shortest TE to maximize T1 shortening properties of ferumoxytol; performed to distinguish lymphatics from small blood vessels

Note.—Contrast-enhanced MR lymphangiography is an off-label use of an MRI contrast agent.

*This technique is commonly used for evaluation of peripheral lymphatics, and this table describes our institutional protocol. With some changes (ie, use of special contrast agents or an adapted technique with the patient performing active movement to move the contrast agent), this technique can also be performed for imaging central lymphatics (19). MRI preparation: intracutaneous GBCA is administered with the patient in the magnet or just before the patient enters (8 mL gadobenate dimeglumine mixed with 2.0 mL 1% lidocaine; 1 mL of mixture administered in each web space of each examined extremity; administration occurs with the patient in the magnet). Optional: slow administration of ferumoxytol intravenous piggyback in an MRI nursing station under nursing observation; 5 mg/kg (up to a maximum dose of 510 mg) diluted in 100 cc of normal saline (administration of ferumoxytol under close clinical observation may be desired given the reported risk of anaphylaxis).

masses with thin septa. Color flow is absent within the cysts, with no or low flow in the septa (Fig 8). On CT images, they are homogeneous fluid-attenuation lesions with thin well-defined walls. On MR images, the multicystic nature and thin septa maybe more easily identified (Fig 9). Enhancement or calcifications in a cystic mass should alert the radiologist to consider a diagnosis other than an LM as these are uncommon findings. At MRI, these lesions are hyperintense on T2-weighted images and iso- to hypointense on T1-weighted images. Fluid-fluid levels are often present (28). Dynamic MR lymphangiography is not usually performed for diagnosis of LMs unless there is an associated lymphatic flow disorder such as can be seen in GLA, as discussed in the next section.

The microcystic type is identified by much smaller cystic structures that permeate through

tissues, often the subcutaneous soft tissues and adjacent muscles (Fig 10). In general, these lesions are harder to treat, with postsurgical recurrence rates as high as 20% and inability to completely resect the entire mass in approximately 30% of cases (29). The mixed type shows imaging overlap of microcystic and macrocystic types with both large and small cysts and typically occurs below the neck. The use of medical therapy including sirolimus for microcystic and mixed types is an area of active investigation (30).

Generalized Lymphatic Anomaly

Formerly known as lymphangiomatosis, GLA is a rare condition in which multiple LMs are identified. It is usually diagnosed in the pediatric population, and the bones are commonly involved (40%–75%) with intramedullary involvement and cortical sparing (31–33). Other

Table 3: Overview of Lymphatic Disorders with Key Imaging Features

Lymphatic Disorders	Imaging Findings (if applicable)
LMs	
Macrocytic malformation	Cysts >1–2 cm often with fluid-fluid levels; typically occurs below the mylohyoid
Microcytic malformation	Small cysts; often subcutaneous or involving muscle
Mixed malformation	Characteristics of both macrocytic and microcytic malformations; typically occur below the neck
Generalized lymphatic anomaly (GLA)	Multiple LMs involving bones and viscera
Gorham-Stout disease (GSD)	Osteolysis affecting multiple bones including the skull, ribs, and spine
Channel-type LM	Chylothorax, chylous ascites, and distention of lymphatic structures
Primary lymphedema	Physical examination findings of lymphedema along with genetic analysis used for diagnosis
Combined vascular malformations	Variable imaging appearance with combination of lymphatic and other vascular structural abnormalities
Acquired lymphatic disorders	
Chyle leaks	Chylous effusions (pericardial, pleural, ascites) Lymphangiography can be used to identify the site of the leak
Pulmonary lymphatic perfusion syndrome	Chylothorax, flow reversal from hilum to lungs at MR lymphangiography
PLE	Progression of contrast agent into the bowel at MR lymphangiography
Secondary lymphedema	Physical examination findings of edema MRI can be used for surgical planning
Miscellaneous lymphatic disorders	
Lymphangiioleiomyomatosis (LAM)	Angiomyolipomas; chylous effusions; uniformly distributed pulmonary cysts; can be associated with tuberous sclerosis

visceral involvement including splenic cysts and macrocytic LMs favor a diagnosis of GLA compared with GSD (34). A subtype of GLA, kaposiform lymphangiomatosis, has a more aggressive course owing to consumptive coagulopathy and thrombocytopenia (35). There is imaging overlap in findings of GLA and kaposiform lymphangiomatosis with those of channel-type LM (36).

Gorham-Stout Disease

GSD, also known as vanishing bone disease, is a rare condition with case reports and series describing its manifestation and attempted treatment. Pathologically, this condition is characterized by lymphatic proliferation in the bones leading to osteolysis (37). The osteolysis at imaging is seen as an area of bone loss that has been reported to affect a wide variety of bones including the skull, spine, and ribs (Fig 11). The condition most commonly occurs in young adults (38). Involvement of the craniofacial bones is more diffuse in patients with GSD compared with GLA (39). The underlying cause is not known. Chylothorax has been associated with this condition and may be a poor prognostic factor (40).

Channel-type LM

The diagnosis of channel-type LM (also known as central conducting lymphatic anomaly) is typically made when there is dysfunction or obstruction of lymphatic flow in the thoracic duct or cisterna chyli (41). Because of the impaired lymphatic drainage, patients are diagnosed with chylothorax, chylous ascites, PLE, and distention of the lymphatic structures, as well as swelling due to lymph seepage (Fig 12). At dynamic lymphatic imaging, impaired central lymphatic flow as well as lymphatic reflux can be seen (42). Distinction of channel-type LM from other lymphatic anomalies is difficult owing to the variability in presentation and overlap in clinical manifestations of these conditions (35).

Acquired Progressive Lymphangioma

Case reports of acquired progressive lymphangioma have most commonly been reported in younger patients who present with a slow-growing erythematous plaque, typically in an extremity (43). This skin manifestation is due to the proliferation of lymphatics in the dermis.

Primary Lymphedema

Primary lymphedema occurs in the setting of impaired lymphatic flow and leads to swelling

Table 4: 2018 ISSVA Classification of Lymphatic Disorders**Simple vascular malformations**

Common cystic LMs
 Macrocystic
 Microcystic
 Mixed cystic
 GLA
 Kaposiform lymphangiomatosis
 LM in GSD
 Channel-type LM
 Acquired progressive lymphatic anomaly
 Primary lymphedema
 Other

Combined vascular malformations

Capillary-lymphatic malformation
 Lymphatic-venous malformation
 Capillary-lymphatic-venous malformation
 Capillary-lymphatic-arteriovenous malformation
 Capillary-lymphatic-venous-arteriovenous malformation

Vascular malformations associated with other anomalies

Klippel-Trenaunay syndrome
 CLOVES syndrome
 Proteus syndrome
 CLAPO syndrome

Note.—CLAPO = capillary vascular malformation of the lower lip, lymphatic malformations of the head and neck, asymmetry, and partial or generalized overgrowth; CLOVES = congenital lipomatous (fatty) overgrowth, vascular malformations, epidermal nevi, and scoliosis or skeletal or spinal anomalies.

of the extremities. There are multiple disease entities that cause primary lymphedema, each described with distinct phenotypic features and genetic alterations (Table 5) (44). One example is Nonne-Milroy disease, an autosomal dominant disorder that manifests shortly after birth with bilateral lower extremity edema, hydroceles and urethral anomalies in men, upward curvature of the toenails, papillomatosis, and prominence of venous structures (45). Mutations of vascular endothelial growth factor receptor-3 have been identified in this disorder (46). Other conditions causing primary lymphedema include lymphedema-distichiasis syndrome, hypotrichosis-lymphedema-telangiectasia syndrome, and lymphedema-choanal atresia. The role of MR lymphangiography in primary lymphedema is currently investigational.

Combined Vascular Malformations

Combinations of vascular malformations that involve lymphatic structures are included as a separate group of disorders in the ISSVA 2018 classification system (4). These conditions are

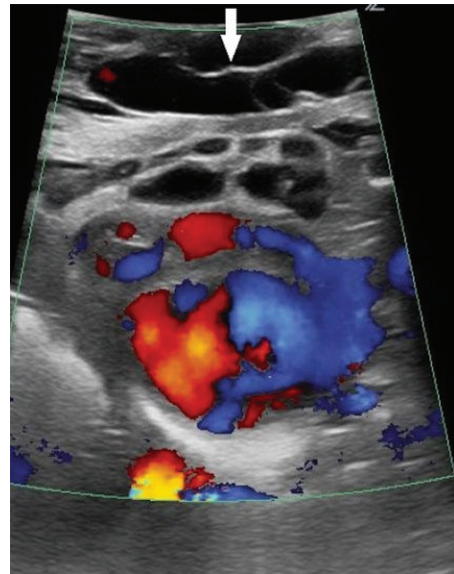


Figure 8. Large LM in a newborn boy that was diagnosed prenatally. Color Doppler US image obtained shortly after birth shows a macrocystic LM in the subcutaneous tissues of the anterior chest wall (arrow) with anechoic cysts and no evidence of internal vascularity. The patient underwent sclerotherapy.

described and named by the vascular components involved; for example a malformation having both capillary and lymphatic components would be named a capillary-lymphatic malformation. These lesions are often divided into high flow, slow flow, or combination, with MRI often playing a role in helping determine the extent and components of these lesion (47). These lesions can occur in conjunction with other anomalies (eg, Klippel-Trenaunay syndrome) (Table 4).

Acquired Lymphatic Disorders and Those Associated with Other Conditions

Chyle Leak

Chyle leaks can be due to disruption or occlusion of abdominal lymphatic channels, cisterna chyli, or lymphatics superior to it. Trauma (most commonly iatrogenic injury from esophageal, heart, or vascular surgery) can lead to chyle accumulation in the thorax, pericardium, or abdominal cavity (Fig 13). Nontraumatic chyle accumulation can occur secondary to obstruction from tumors or congenital pathologic conditions (Fig 14). A special form of nontraumatic chyle accumulation occurs in the setting of pulmonary lymphatic perfusion syndrome, which is discussed in the following section.

There is limited evidence regarding optimal management of chyle leaks, but options include

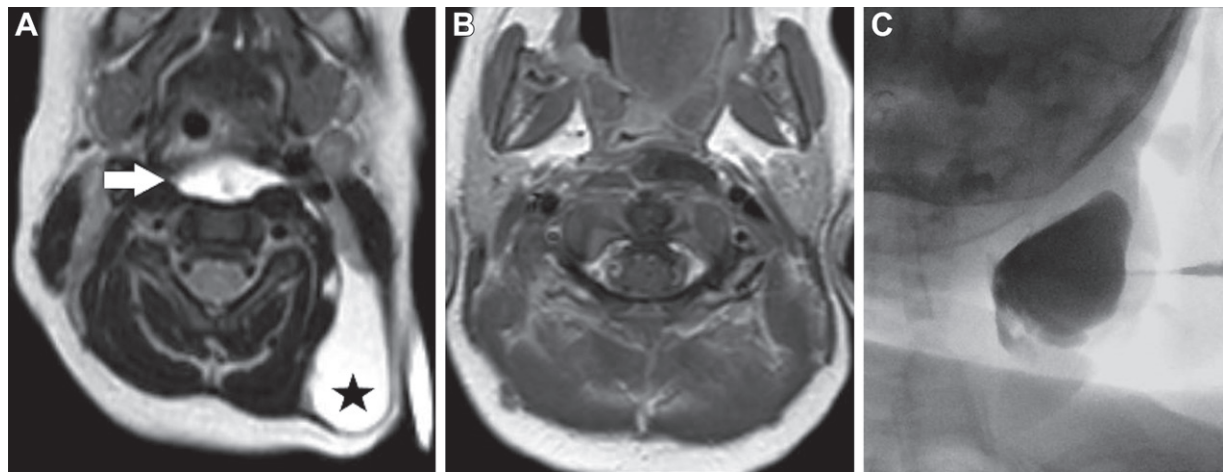


Figure 9. Macroscopic LM in a 16-month-old boy who presented with a neck mass. (A) Axial T2-weighted MR image shows a hyperintense mass that invaginates through multiple tissue planes involving the posterior neck (★) as well as the prevertebral tissues (arrow). (B, C) On an axial contrast-enhanced T1-weighted MR image (B), no enhancement is seen. The imaging features are consistent with a macrocystic LM, for which sclerotherapy was performed under fluoroscopic guidance (C).

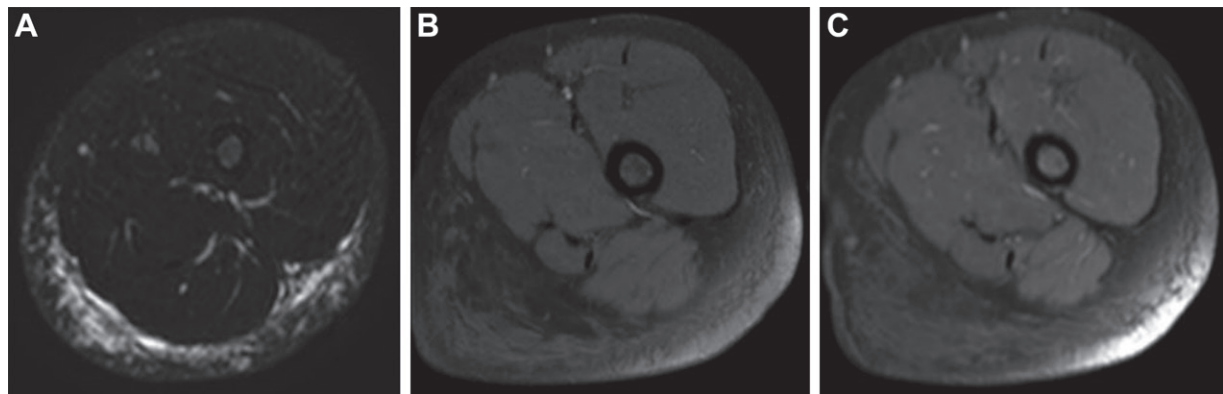


Figure 10. Microcystic LM in a 15-year-old adolescent girl with a leg mass. (A) Axial T2-weighted MR image of the lower extremity shows multiple tiny hyperintensities in the posterior thigh that permeate through the soft tissues. (B, C) When comparing axial T1-weighted noncontrast (B) and contrast-enhanced (C) images, no enhancement is seen. The findings were due to a microcystic LM.

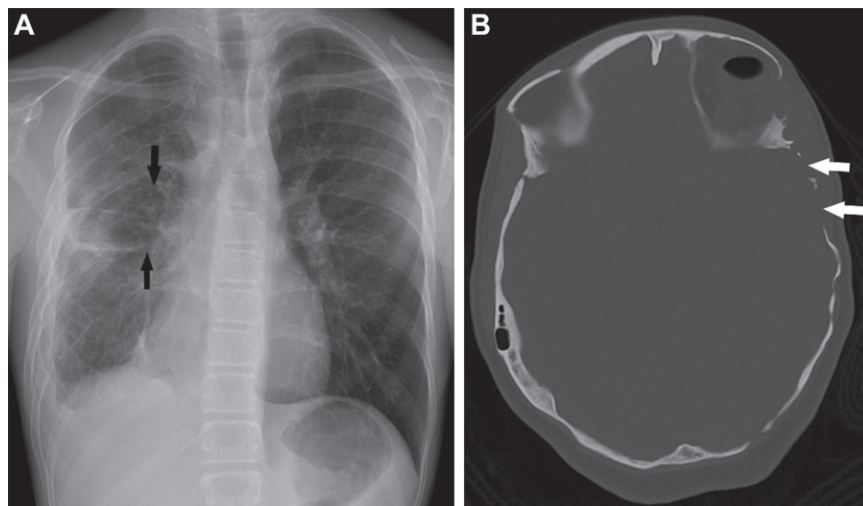


Figure 11. GSD in a 30-year-old man with osteolysis depicted on a chest radiograph (A) in the ribs (arrows in A) and on an axial noncontrast CT image (B) of the head (arrows in B).

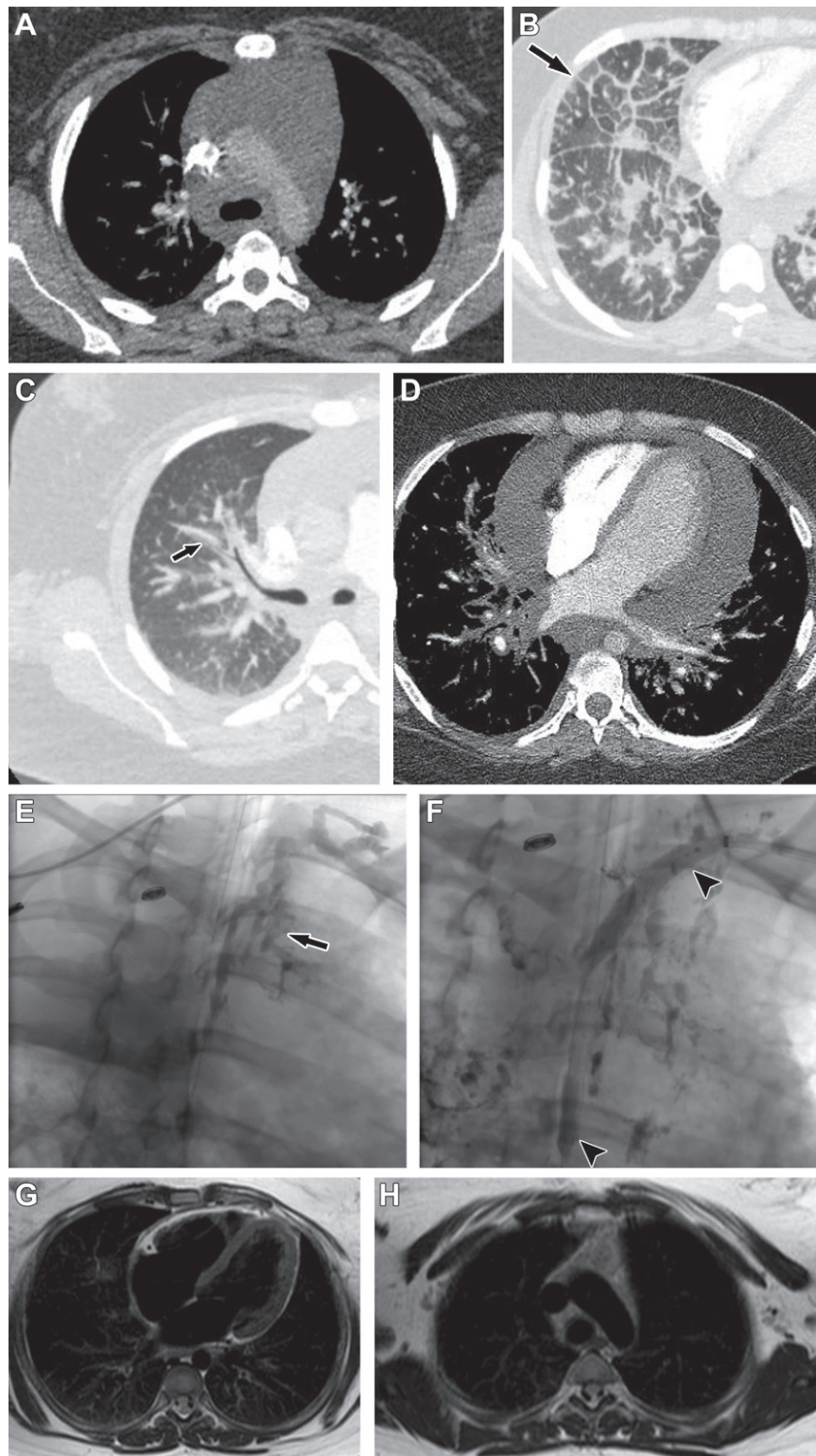


Figure 12. Channel-type LM in a 24-year-old woman who presented with a mediastinal mass. (A) Axial contrast-enhanced CT image shows fluid attenuation that insinuates throughout the mediastinum. (B, C) Axial contrast-enhanced CT images show thickening along the interlobular septa as well as the bronchovascular structures, highlighting the dilatation of the lymphatic vessels in the lungs (arrow). (D) Axial contrast-enhanced CT image shows a chylous pericardial effusion causing tamponade that required pericardiocentesis. (E–H) Thoracic duct lymphangiographic images (E, F) show abnormal contrast agent flow (arrow in E), which was covered by a stent (arrowheads outline the cranial and caudal aspects of the stent in F), with resolution of the pericardial effusion (G) and significant decrease in the mediastinal mass (H) on axial black-blood T2-weighted MR images (G, H). Imaging findings were suggestive of channel-type LM.

conservative treatment, interventional procedures, and surgery. Conservative therapy includes dietary changes to decrease chyle production (modified medium-chain triglyceride or fat-free diet), total parenteral nutrition, or medical therapy including somatostatin analogs (48). Interventional radiology plays a key role in treatment

by thoracic duct intervention, which is described in the “Therapeutics” section. Surgical techniques including thoracic duct ligation (with or without pleurodesis in cases of chylothorax) and pleuroperitoneal shunts are also possible, with therapeutic decisions based on patient-specific factors and institutional expertise (49,50).

Table 5: Causes of Primary Lymphedema

Type of Primary Lymphedema	Defining Characteristics	Gene
Nonne-Milroy syndrome	Lower extremity edema, hydroceles, male urethral anomalies, upward curvature of the toenails, papillomatosis, and prominence of venous structures	Vascular endothelial growth factor receptor-3
Primary hereditary lymphedema	Bilateral lymphedema	Vascular endothelial growth factor C Gap junction protein γ
Lymphedema-distichiasis	Late-onset lymphedema, abnormality of the eyelashes, and ptosis	Forkhead box c2
Hypotrichosis-lymphedema-telangiectasia	Lymphedema, alopecia, and telangiectasia	SRY-box transcription factor 18
Primary lymphedema with myelodysplasia	Lymphedema of the lower extremities and genitals, immune dysfunction, skin warts, and deafness	<i>GATA2</i> , transcription factor
Primary GLA	Lymphedema of the face and lower extremities, intestinal lymphangiectasia, intellectual deficits, and facial anomalies	Collagen and calcium-binding epidermal growth factor (EGF)
Microcephaly with or without chorioretinopathy, lymphedema, or mental retardation	Lower extremity edema with microcephaly and mental retardation	Kinesin family member 11
Lymphedema-choanal atresia	Bilateral choanal atresia and lower extremity edema	Protein tyrosine phosphatase non-receptor type 14

Pulmonary Lymphatic Perfusion Syndrome

Pulmonary lymphatic perfusion syndrome occurs when abnormal lymphatic flow develops in the lungs or pleural space. This can result in the development of a chylothorax and/or plastic bronchitis. While chylothorax is diagnosed by the results of pleural fluid analysis, plastic bronchitis is diagnosed by the removal (either by self-expectoration or bronchoscopy) of branching casts made of cross-linked mucin (51). This condition has been described in congenital heart disease, specifically in patients with a single ventricle after Fontan palliation, where flow reversal is due to increased central venous pressures secondary to Fontan physiology (52). MR lymphangiography can be performed to demonstrate the abnormal lymphatic flow (Fig 15). To review, in the normal state lymphatic flow occurs from the lungs to the hilum and then into the central veins. However, when there is impaired flow, whether that be functional or anatomic impedance to flow, this pattern of flow reverses. Specifically, flow goes from the hilum into the lungs (Fig 16), which can be seen at dynamic imaging of the lymphatics.

Protein-losing Enteropathy

PLE can be another manifestation of impaired lymphatic function due to either gastrointestinal causes (eg, primary intestinal lymphangiectasia) or cardiac causes leading to elevated central

venous pressure (such as in patients with congestive cardiac failure) (53). This is a well-described although uncommon complication of the Fontan procedure and contributes to an increase in mortality and morbidity in these patients (54). Patients present with chronic diarrhea and peripheral edema, which are due to the loss of proteins through the duodenum through abnormal hepatoduodenal lymphatic communications (55,56). The abnormal lymphatic flow can be demonstrated at MR lymphangiography by the progression of lymphatically injected contrast agents into the bowel, which is best observed during dynamic imaging to assess contrast agent movement during peristalsis (Fig 17).

Secondary Lymphedema

Surgical or radiation therapy-related chronic inflammation and scarring in an extremity leads to disruption of lymphatic channels and subsequently to abnormalities in lymph drainage. Secondary lymphedema is an under-recognized disease with an estimated 200 million people worldwide with this condition (44). In the United States, lymph node dissection and radiation therapy, used for the treatment of malignancy, is the most common cause of lymphedema. The skin may ulcerate and recurrent infections can occur. Traditionally, first-line approaches for treatment included skin care, manual lymphatic

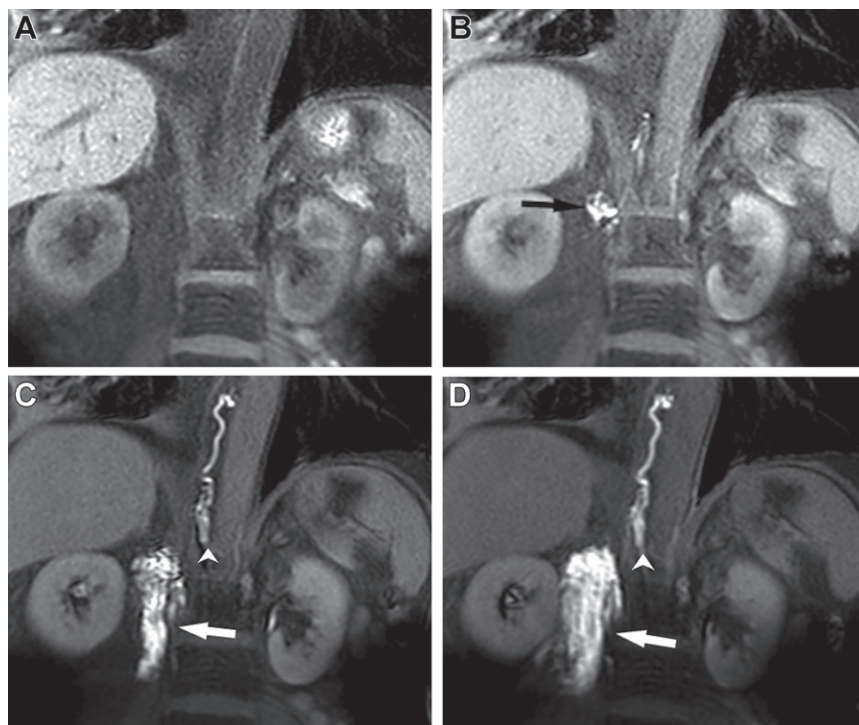


Figure 13. Chyle leak in a 47-year-old woman with chylous pleural effusion and ascites. MRI was performed to assess for chyle leak. Coronal noncontrast (A) and dynamic contrast-enhanced (B–D) 3D T1-weighted GRE images with fat saturation after intranodal administration of gadobenate dimeglumine show a chyle leak into the right retroperitoneum (arrow in B–D). Note the good enhancement within the thoracic duct (arrowhead in C and D).

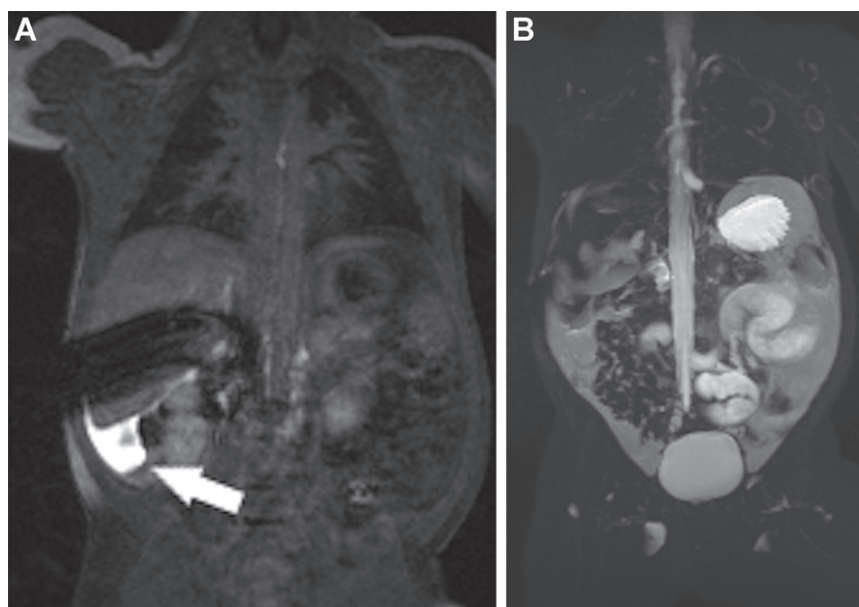


Figure 14. Chyle leak in an 8-month-old boy with a history of chylous ascites. (A) Coronal MR lymphangiographic image after injection of intrahepatic contrast agent shows contrast agent extravasation into the right side of the peritoneal cavity (arrow). (B) On a coronal T2-weighted MR image, the severity of the chylous ascites is better appreciated.

drainage with massage, exercise, and compression devices. However, in recent years, surgical treatments such as lymphovenous anastomosis (LVA) and vascularized lymph node transplant (VLNT) have been used to manage lymphedema by

restoring lymphatic drainage to an affected limb (57). In LVA, small lymphatics peripheral to a site of lymphatic obstruction are anastomosed to low-pressure venules resulting in venous outflow of lymph from an extremity (58).

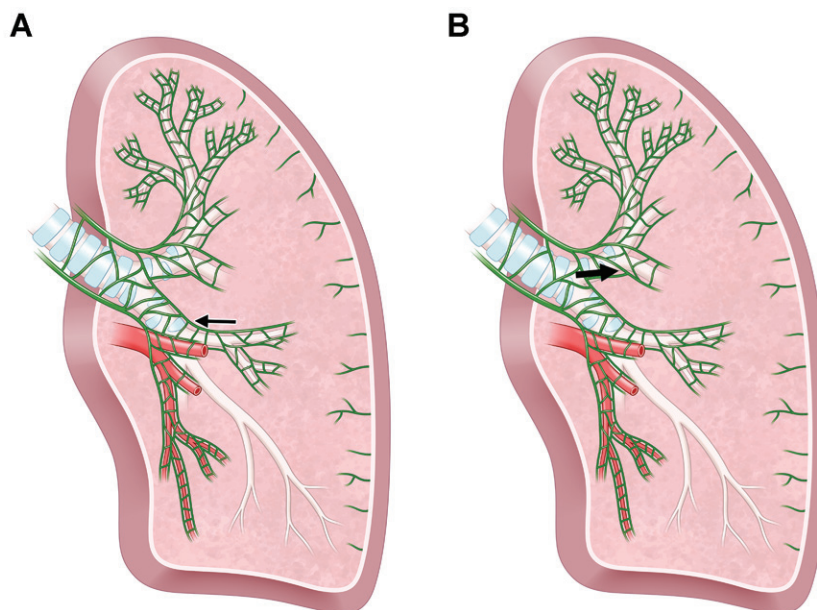


Figure 15. Illustrations show normal and abnormal lymphatic flow. Normal lymphatic flow (arrow in **A**) in the lungs moves toward the hilum. When there are lymphatic flow abnormalities, flow may be reversed, with lymphatic fluid (arrow in **B**) moving away from the hilum to the lungs.

As these surgical procedures are becoming more commonplace, so is MR lymphangiography of the peripheral extremities. In this setting, MRI is used for patient selection and surgical planning, not for lymphedema diagnosis (Fig 18). Specifically, dermal backflow on MR images may indicate a higher likelihood of benefit for reduction of lymphedema from LVA (Fig 18C). Scientific evidence supporting the use of MRI for this is currently limited. Nevertheless, many microsurgeons consider it helpful and complementary to other preoperative imaging modalities such as lymphoscintigraphy (59).

Lymphangioleiomyomatosis

Some lymphatic disorders are not covered in any of the categories previously discussed. One of these (LAM) is discussed in this section.

There are a variety of manifestations of LAM, including pulmonary cysts and renal angiomyolipomas. Two forms exist, sporadic and tuberous sclerosis complex (TSC)-associated forms, both of which are due to mutations in one of the tuberous sclerosis genes (*TSC1* and *TSC2*); a mutation in the *TSC2* gene is the most common (60). The condition occurs predominately in women. The mutation allows the proliferation of LAM cells, which are considered a low-grade metastasizing neoplasm that spreads via the lymphatics (61). Diagnostic guidelines outline that in addition to the characteristic findings of thin uniformly distributed pulmonary cysts, an additional one or more feature(s) are required to make the

diagnosis, including (*a*) an underlying diagnosis of TSC, (*b*) angiomyolipomas, (*c*) chylous effusion, (*d*) elevated vascular endothelial growth factor (VEGF)-D level greater than or equal to 800 pg/mL, and/or (*e*) lymphangioleiomyomas (Fig 19) (62). Clinical practice guidelines recommend treatment with sirolimus if there is declining or abnormal lung function (63).

Therapeutics

Interventional radiology plays an integral part in the treatment of both acquired and inherited lymphatic abnormalities. Several approaches may be used depending on the clinical scenario and underlying pathologic condition, including selective embolization, percutaneous or catheter-based sclerosis, or more novel techniques such as thoracic duct stent placement.

Thoracic Duct Embolization

Thoracic duct embolization is an effective minimally invasive procedure for treatment of chylothorax and other chyle leaks. Although most cases of chyle leak occur in the setting of trauma or surgery, this technique has also shown efficacy in idiopathic nontraumatic causes where a leak might not be identified at lymphangiography. Thoracic duct embolization has a reported success rate greater than 60% for effective thoracic duct embolization, with a low risk of major complication (<2%) (64). Multiple techniques have been described to obtain access to the thoracic duct (65).

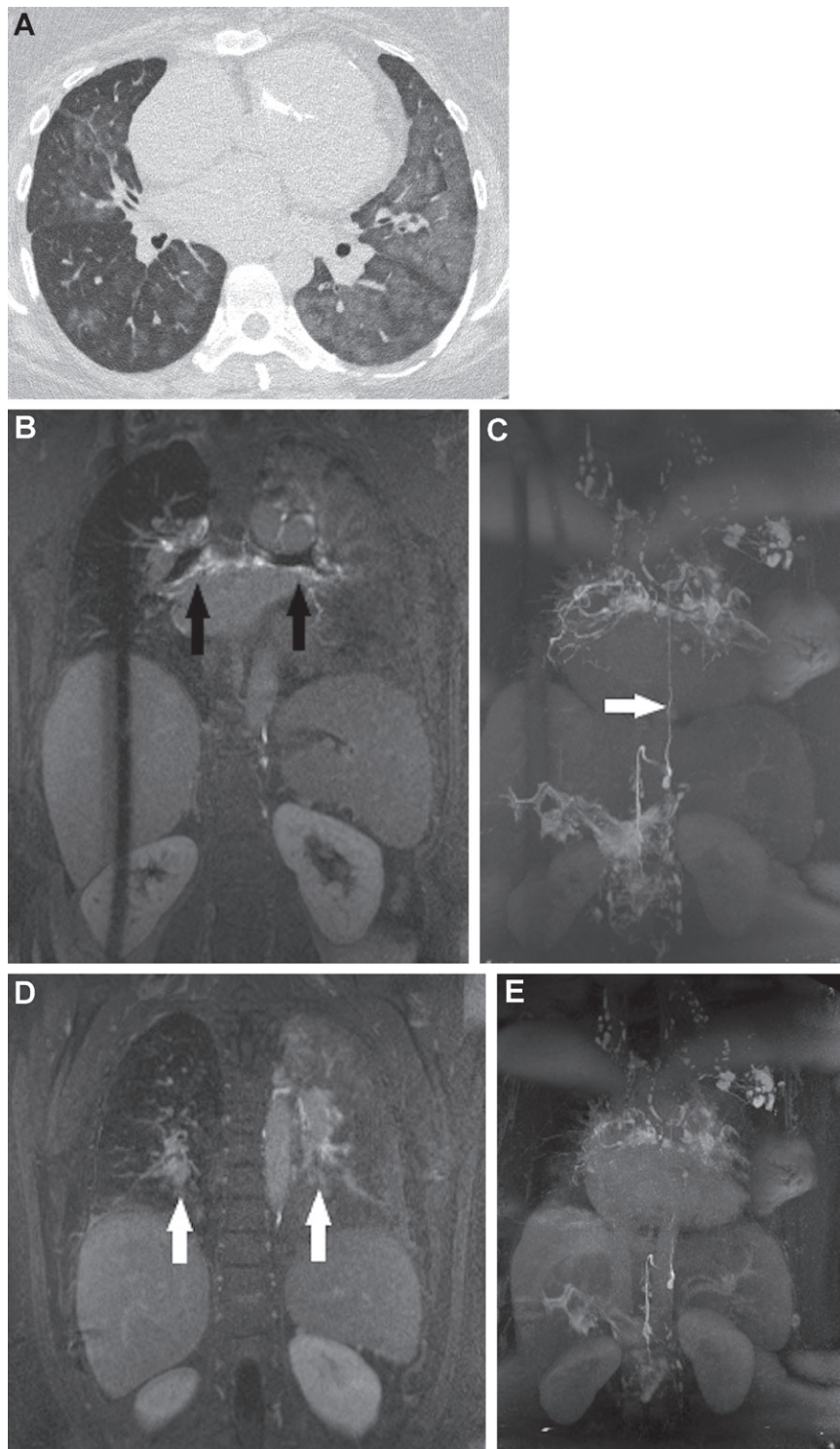


Figure 16. Pulmonary perfusion syndrome in a 47-year-old woman with a history of tetralogy of Fallot after repair. (A) Axial noncontrast CT image of the chest shows bilateral centrilobular ground-glass opacities. (B–D) MR lymphangiography was performed using both intrahepatic (B, coronal contrast-enhanced T1-weighted image; C, maximum intensity projection [MIP] image) and intranodal (D, coronal contrast-enhanced T1-weighted image; E, MIP image) injections. Bilateral hilar perfusion is seen with both injection techniques (arrows in B and D), a finding of pulmonary perfusion syndrome. Note that with intrahepatic injection, the entire thoracic duct can be seen (arrows in C).

Once the cisterna chyli is opacified with a contrast agent, it is directly accessed typically from an anterior approach with a small-gauge needle. The cisterna chyli and thoracic duct are next cannulated with a microcatheter, and embolization of the thoracic duct through the microcatheter is performed (Fig 20). Embolization is most commonly performed with microcoils and/or cyanoacrylate glue (66). Overall, patients do well after the

procedure, with delayed complications including chronic diarrhea as well as leg and abdominal swelling, which are usually well tolerated (67).

Sclerosis of LMs

Percutaneous sclerotherapy can be used in the treatment of LMs with success rates higher for macrocystic LMs compared with those for microcystic and mixed LMs (68). Treatment of

Figure 17. PLE after thoracic duct ligation in a 21-year-old man with a history of dextrocardia with transposition of the great vessels, right atrioventricular valve, and pulmonary atresia after staged palliation culminating in a fenestrated extracardiac Fontan procedure. Coronal contrast-enhanced MR lymphangiographic images after intrahepatic injection of a contrast agent show leakage of the contrast agent intraluminally into the duodenum (arrow in A) with progression of the contrast agent within the bowel (arrow in B).

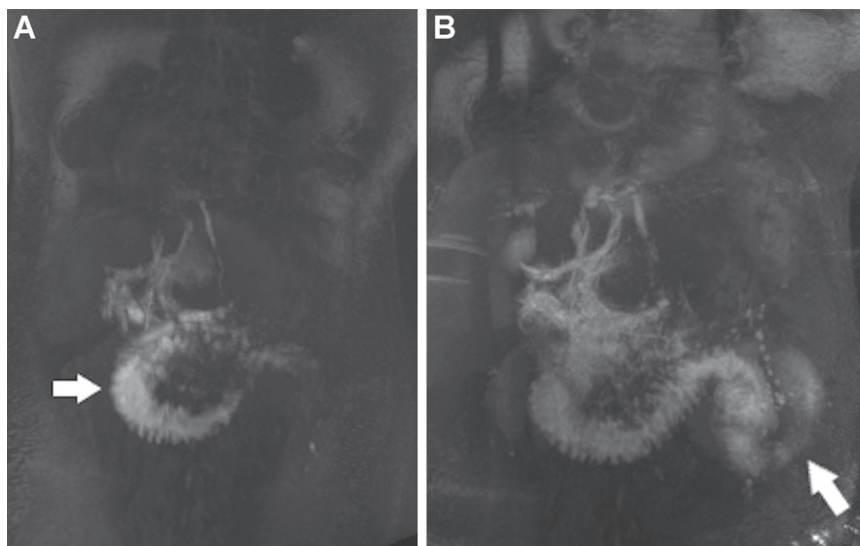
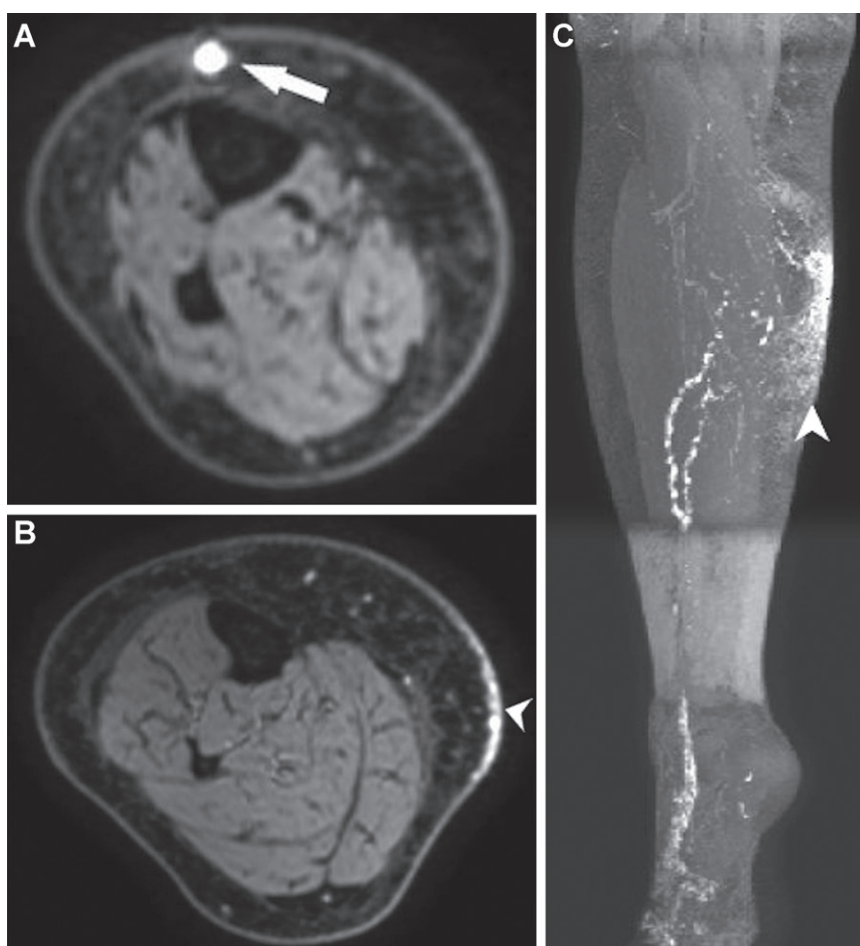


Figure 18. Dermal backflow in a 55-year-old woman with a history of ovarian cancer and pelvic lymphadenectomy who presented with right leg edema. Axial 3D T1-weighted GRE MR images of the mid (A) and upper (B) calf and MIP image of the right lower extremity (C) after intracutaneous administration of a contrast agent into the web spaces of the right foot show a large lymphatic channel ascending the anterior calf (arrow in A) and dermal backflow to the medial upper calf (arrowhead in B and C). Dermal backflow is intermittently identified at MRI in some patients with secondary lymphedema. Of note, ferumoxytol was administered to this patient to null signal from intravascular uptake of gadolinium before image acquisition.



macrocystic LMs with sclerotherapy has similar outcomes to those of surgical therapy. Surgery is therefore reserved for cases where sclerotherapy fails or urgent intervention is needed owing to the immediate threat of death or significant morbidity given the location, such as lesions causing mass effect on the airway (69).

A large variety of sclerosing agents have been used including picibanil (OK-432), ethanol, bleomycin, and doxycycline. There is lack of evidence supporting the use of one agent over another (70). By using imaging guidance (US, CT, fluoroscopy, and/or MRI), the LM is accessed and treated with a variety of injection



Figure 19. LAM in a 45-year-old woman with cystic lung disease. Coronal CT image shows diffuse pulmonary cysts. Hepatic and renal fat-containing masses (angio-myolipomas) were also seen (not shown).

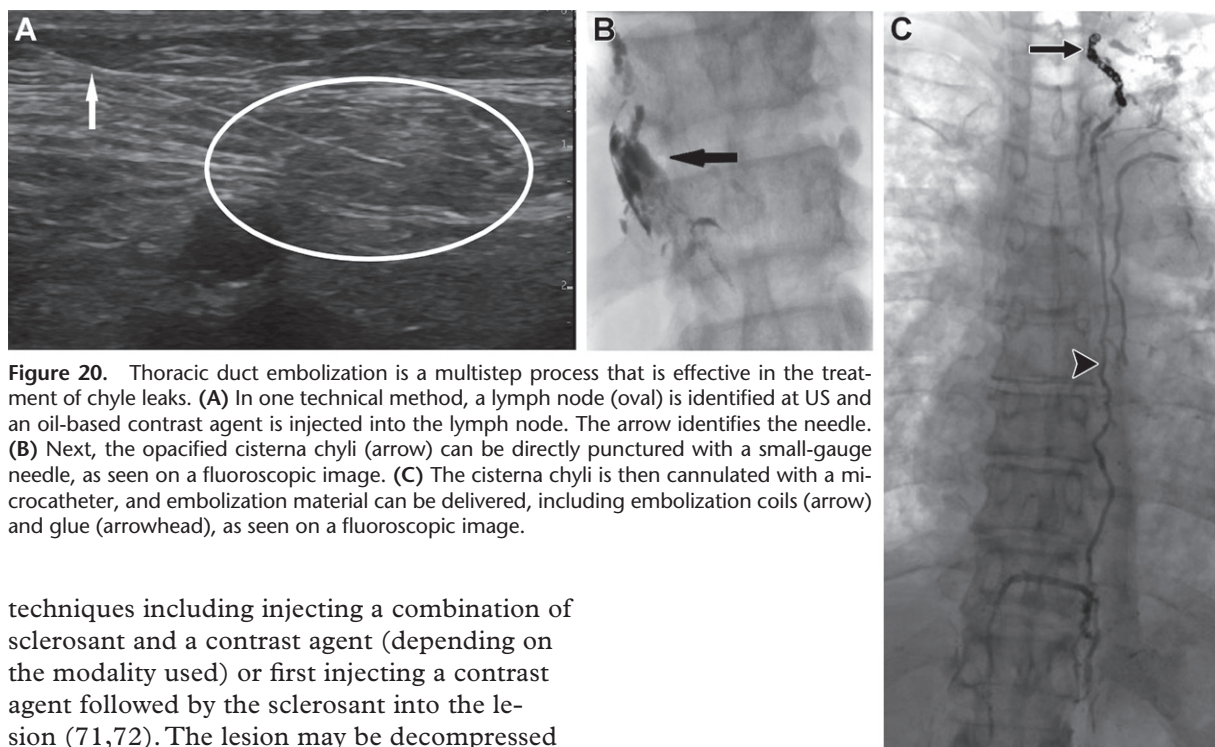


Figure 20. Thoracic duct embolization is a multistep process that is effective in the treatment of chyle leaks. (A) In one technical method, a lymph node (oval) is identified at US and an oil-based contrast agent is injected into the lymph node. The arrow identifies the needle. (B) Next, the opacified cisterna chyli (arrow) can be directly punctured with a small-gauge needle, as seen on a fluoroscopic image. (C) The cisterna chyli is then cannulated with a microcatheter, and embolization material can be delivered, including embolization coils (arrow) and glue (arrowhead), as seen on a fluoroscopic image.

techniques including injecting a combination of sclerosant and a contrast agent (depending on the modality used) or first injecting a contrast agent followed by the sclerosant into the lesion (71,72). The lesion may be decompressed by aspirating the fluid within it before or after introduction of the sclerosant. Real-time imaging during the procedure may be performed to assess the success of the procedure and monitor for extravasation. Preprocedural antibiotic therapies are typically given and steroid therapy may also be administered to reduce postprocedural swelling (73). Postprocedural side effects of sclerosis include fever and injection-site swelling and erythema (73). Multiple treatments with sclerotherapy may be required for larger lesions.

Thoracic Duct Stent Placement

There is renewed interest in the use of stent placement for treatment of chylous effusions and other disorders that lead to chyle accumula-

tion. Case reports and small series of patients have been described in whom stent grafts have been placed in the thoracic duct at a site of impaired drainage or leakage (74). Through the use of stent grafts rather than embolization, it is hypothesized that continued patency of the central lymphatics can avoid delayed complications including lymphedema or PLE.

Conclusion

The complex anatomic and physiologic role of the lymphatic system continues to be an area of evolving knowledge. Imaging plays a central role in improving our understanding of this

organ system and in diagnosis and treatment of lymphatic abnormalities. It is important for radiologists to be aware of the normal and variant lymphatic anatomy, disorders of the lymphatic system, and evolving diagnostic techniques and treatment options.

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